

PHASE-WISE BREAKDOWN: JUNE & JULY

Your 61-Day Summer Challenge Detailed Plan

TWO-PHASE STRUCTURE

PHASE 1: JUNE (30 Days) - BUILDING FOUNDATION
Days 1-30
Focus: Learn fundamentals, build locally, understand
Output: Working pipeline on laptop

PHASE 2: JULY (31 Days) - MASTERY & DEPLOYMENT
Days 31-61
Focus: Deploy, optimize, polish, publish
Output: Production pipeline, blog, portfolio live

PHASE 1: JUNE (Days 1-30) - BUILDING FOUNDATION

PHASE 1 OVERVIEW

Goal: Build a solid foundation across all 4 tracks **Output:** Working Spark pipeline on your laptop **Duration:** 30 days **Total Hours:** 180 hours (6 hrs/day × 30 days)

Success Criteria:

- ✅ Spark pipeline working locally
- ✅ 30 DSA problems solved
- ✅ Python fundamentals strong
- ✅ Basic website structure
- ✅ 30 daily GitHub commits

PHASE 1 - WEEK BY WEEK

WEEK 1 (June 1-7): SET THE STAGE

Week Goal: Set up all accounts, tools, and fundamentals

Daily Breakdown:

DAY 1 (June 1) - LAUNCH

PYTHON (30m)

- Print "Hello World"
- Variables & data types
- First program

TECH (30m)

- Join r/learnprogramming
- Follow @TechCrunch
- Read 1 tech article

DSA (90m)

- Watch: Java setup & first program
- Write HelloWorld.java
- Understand Java basics

Problem Count: 1/35

DATA ENG (120m)

- Create AWS free tier account
- Launch EC2 t2.micro instance
- Understand AWS console

AWS Status:  Ready

WEB (90m)

- Learn HTML structure
- Create index.html
- Basic page structure

COMMIT

Message: "Day 1: AWS account created, Java hello world, HTML basics"

DAY 2 (June 2)

PYTHON (30m)

- Variables exercise
- Simple calculations
- Print statements

TECH (30m)

- Read: "What is Cloud Computing"
- Follow: 1 AWS account
- Bookmark tech sites

DSA (90m)

- Watch: Arrays in Java
- Create ArrayExamples.java
- Code 2-3 array programs

Problem Count: 3/35

DATA ENG (120m)

- Watch: EC2 tutorial
- SSH into EC2 instance
- Basic Linux commands (ls, cd, pwd)

AWS Status:  SSH working

WEB (90m)

- Learn CSS basics
- Create style.css
- Basic styling

COMMIT

Message: "Day 2: EC2 SSH working, Java arrays, CSS added"

DAY 3 (June 3) - PYTHON DEEP

PYTHON (30m)

- Loops (for, while)
- Write 3 programs with loops

DSA (90m)

- Solve 2-3 LeetCode Easy
- Array problems

Problem Count: 5/35

DATA ENG (120m)

- Install Python 3
- pip install requirements
- Write first Python script

WEB (90m)

- Flexbox introduction
- Responsive design basics

COMMIT

Message: "Day 3: Python loops, DSA 5 problems, basic responsive"

DAY 4 (June 4) - DSA FOCUS

PYTHON (30m)

- Conditional statements

- if/else practice

DSA (90m)

- Watch: String problems
- Solve 2 string problems

Problem Count: 8/35

DATA ENG (120m)

- Install Jupyter Notebook
- Create first notebook
- Pandas introduction

WEB (90m)

- CSS Grid basics
- Build 2-column layout

COMMIT

Message: "Day 4: Pandas installed, Jupyter ready, CSS grid"

DAY 5 (June 5)

PYTHON (30m)

- Functions in Python
- Write 3 functions

DSA (90m)

- More string problems
- Two-pointer technique

Problem Count: 10/35

DATA ENG (120m)

- Read CSV file in Pandas
- Explore dataset
- Basic statistics

WEB (90m)

- JavaScript basics
- Variables & operators
- Simple scripts

COMMIT

Message: "Day 5: 10 DSA problems! Pandas CSV reading working"

DAY 6 (June 6)

PYTHON (30m)

- Lists & dictionaries
- Data structure practice

DSA (90m)

- Binary search problems
- Searching algorithms

Problem Count: 12/35

DATA ENG (120m)

- Data cleaning in Pandas
- Handle missing values
- Remove duplicates

WEB (90m)

- DOM manipulation
- Select & change elements
- Event listeners intro

COMMIT

Message: "Day 6: Data cleaning working, JavaScript DOM basics"

DAY 7 (June 7) - WEEK 1 REVIEW

PYTHON (30m)

- Review week 1
- Practice challenges

DSA (90m)

- Solve harder problems
- Review arrays & strings

Problem Count: 15/35

DATA ENG (120m)

- Combine Pandas operations
- Create small pipeline
- Save output

WEB (90m)

- Responsive website check
- Mobile testing
- Fix issues

COMMIT

Message: "Day 7: Week 1 complete! 15 DSA problems, Pandas pipeline working"

WEEK 1 CHECKPOINT:

- ☐ Python fundamentals clear

□ Python + Pandas + AWS working
□ 15 DSA problems solved
□ AWS + Python + Pandas working
□ Responsive website
□ 7 daily commits
Status:  FOUNDATION LAID

WEEK 2 (June 8-14): SPARK BEGINS 🔥

Week Goal: Spark fundamentals, first data transformation

Key Milestones:

- Day 8: PySpark installed
- Day 10: Read CSV in Spark
- Day 12: Transformations working
- Day 14: First complete mini-pipeline

Daily Pattern:

Days 8-9: PySpark Setup

- PySpark installation
- SparkSession creation
- DataFrames introduction

Days 10-11: Read Data

- Read CSV files
- Show data
- Schema understanding

Days 12-13: Transformations

- select() operation
- filter() operation
- groupBy() aggregations

Day 14: Review & Consolidate

- Combine all operations
- Create mini-pipeline
- Document code

WEEK 2 CHECKPOINT (Day 14):

✅ Achievements:

- ❑ 20 DSA problems solved
- ❑ PySpark working
- ❑ CSV reading in Spark ✅
- ❑ Basic transformations ✅
- ❑ Website interactive

Status: 🟢 MOMENTUM BUILDING

Total: 84 hours completed

WEEK 3 (June 15-21): BUILD YOUR PIPELINE 💪

Week Goal: Complete working Spark pipeline

What You'll Build: Sales Analysis Pipeline

- Input: orders.csv
- Process: Data cleaning, transformations, aggregations
- Output: Sales by status, top products
- Outcome: Professional data engineering project

Key Milestones:

- Day 15: Understand data, plan pipeline
- Day 17: Data cleaning working
- Day 19: Complex transformations
- Day 21: Save results + documentation

Pipeline Steps:

1. Read orders.csv
2. Data type casting
3. Remove invalid records
4. Calculate metrics
5. Group and aggregate
6. Save to CSV

WEEK 3 CHECKPOINT (Day 21):

✅ Achievements:

- ❑ 25 DSA problems solved
- ❑ Spark pipeline WORKING ✅
- ❑ Data cleaning implemented

- Data cleaning implemented
- Aggregations complete
- Output files saved

Status:  PROJECT TAKING SHAPE
Total: 126 hours completed

WEEK 4 (June 22-30): DOCKER & DOCUMENTATION 📦

Week Goal: Prepare for deployment, document everything

What You'll Do:

- Clean up code
- Add comments
- Create requirements.txt
- Write Dockerfile
- Build Docker image locally

Dockerfile Preview:

```
dockerfile

FROM python:3.9-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install -r requirements.txt

COPY . .

CMD ["python", "sales_pipeline.py"]
```

Key Milestones:

- Day 22-23: Code cleanup & comments
- Day 24-25: Create Dockerfile
- Day 26: Build Docker image
- Day 27: Test Docker image
- Day 28-29: Push to Docker Hub
- Day 30: Final polish

JUNE 30 CHECKPOINT - PHASE 1 COMPLETE:

✅ PHASE 1 ACHIEVEMENTS:

- ❑ 30 DSA problems solved ✅
- ❑ Spark pipeline WORKING ✅
- ❑ Docker image CREATED ✅
- ❑ Pushed to Docker Hub ✅
- ❑ Professional documentation ✅
- ❑ Portfolio website responsive ✅
- ❑ 30 daily GitHub commits ✅

STATISTICS:

- Total hours: 180
- Days completed: 30
- Problems solved: 30
- Python level: Intermediate
- Spark level: Beginner+

WHAT YOU HAVE:

- Working Spark pipeline on laptop
- Docker image on Docker Hub
- Professional GitHub repository
- Responsive website structure
- 30 days of consistent learning

Status: 🟢 READY FOR PHASE 2

⚡ PHASE 2: JULY (Days 31-61) - MASTERY & DEPLOYMENT

📌 PHASE 2 OVERVIEW

Goal: Deploy to production, master all skills, publish results **Output:** Live Spark pipeline on AWS, portfolio website live, blog published **Duration:** 31 days **Total Hours:** 186 hours (6 hrs/day × 31 days)

Success Criteria:

- ✅ Pipeline deployed on AWS EC2
 - ✅ Docker container running on AWS
 - ✅ 50+ DSA problems solved
 - ✅ Portfolio website live on GitHub Pages
 - ✅ Blog post published on Medium
 - ✅ 61 daily GitHub commits total
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WEEK 5 (July 1-7): DEPLOY TO AWS 🚀

Week Goal: Get pipeline running on AWS EC2

What You'll Learn:

- Docker Hub workflow
- AWS EC2 best practices
- Container orchestration basics

Key Milestones:

- Day 31: Docker Hub image pushed
- Day 33: AWS EC2 instance ready
- Day 35: Docker installed on EC2
- Day 37: Container running on AWS
- Day 39: Everything tested & working

Daily Breakdown:

Days 31-32: Docker Hub

- Create Docker Hub account
- Push image to Docker Hub
- Verify image online

Days 33-34: AWS EC2 Setup

- Create new EC2 instance
- Configure security groups
- Update system packages

Days 35-36: Docker on EC2

- Install Docker on EC2
- Pull image from Docker Hub
- Test image locally

Days 37-39: Run Pipeline

- Start container
- Map volumes
- Verify output
- Set up cron job for scheduling

WEEK 5 CHECKPOINT (Day 39):

- ✔ Achievements:
- ☐ 35 DSA problems solved ✔
 - ☐ Docker Hub image LIVE ✔
 - ☐ AWS EC2 READY ✔
 - ☐ Container RUNNING ✔
 - ☐ Pipeline DEPLOYED ✔

Status: ● PRODUCTION LIVE!
Total: 234 hours completed

WEEK 6 (July 8-14): MASTERY & OPTIMIZATION ⚡

Week Goal: Optimize performance, add monitoring

What You'll Do:

- Performance tuning
- Add logging
- Setup monitoring
- Website deployment
- DSA deep practice

Key Milestones:

- Day 40: Website deployed on GitHub Pages
- Day 42: Pipeline performance optimized
- Day 44: Monitoring setup
- Day 46: Caching implemented
- Day 48: Error handling complete

Daily Breakdown:

Days 40-42: Website Deployment

- Deploy to GitHub Pages
- Domain setup
- Mobile testing
- SEO optimization

Days 43-44: Performance

Speed optimization

- Spark optimization
- Query optimization
- Cache implementation

Days 45-46: Monitoring

- AWS CloudWatch setup
- Log aggregation
- Alert configuration

Days 47-48: Error Handling

- Try-catch blocks
- Graceful failures
- Retry logic

WEEK 6 CHECKPOINT (Day 48):

- ✅ Achievements:
- ❑ 40 DSA problems solved ✅
 - ❑ Website LIVE on GitHub Pages ✅
 - ❑ Pipeline OPTIMIZED ✅
 - ❑ Monitoring SETUP ✅
 - ❑ Professional deployment ✅

Status: 🟢 INTERVIEW READY!

Total: 288 hours completed

WEEK 7 (July 15-21): BLOG & DOCUMENTATION 📝

Week Goal: Write blog post, complete all documentation

What You'll Write:

- Technical blog post on Medium
- Installation guide
- API documentation
- Architecture diagram
- README files

Blog Post Outline:

1. Introduction (Why I built this)
2. Tech stack explanation

3. Pipeline walkthrough
4. Challenges faced
5. Key learnings
6. Deployment process
7. Results & performance
8. Future improvements

Key Milestones:

- Day 49: Blog outline ready
- Day 51: Blog draft complete
- Day 53: Blog published on Medium
- Day 54: Installation guide ready
- Day 56: Architecture documented
- Day 57: All docs complete

WEEK 7 CHECKPOINT (Day 57):

- ✓ Achievements:
- 45 DSA problems solved ✓
 - Blog post PUBLISHED ✓
 - Installation guide READY ✓
 - Architecture DOCUMENTED ✓
 - Professional portfolio ✓

Status: ● SHOWCASE READY!

Total: 342 hours completed

WEEK 8 (July 22-31): FINAL POLISH & AUG 1 🏆

Week Goal: Final review, self-assessment, prepare for interviews

What You'll Do:

- Code review & refactor
- Self DSA assessment test
- Final documentation
- Portfolio polish
- Preparation for interviews

Key Milestones:

- Day 58: DSA self-test (40+ problems)

- Day 59: Code refactoring
- Day 60: Final documentation
- Day 61: Portfolio COMPLETE!

Self Assessment Test:

Total Problems Solved: 50+

- ✓ Arrays mastery
- ✓ Strings mastery
- ✓ Linked Lists basics
- ✓ Stacks & Queues basics
- ✓ Trees intro

Time Complexity: Can explain $O(n)$, $O(\log n)$, etc.

Space Complexity: Understand trade-offs

Interview Ready: YES! ✓

JULY 31 / AUG 1 FINAL CHECKPOINT:

✓ COMPLETE PHASE 2 ACHIEVEMENTS:

DATA ENGINEERING:

- Spark pipeline DEPLOYED ✓
- Docker PRODUCTION READY ✓
- AWS OPTIMIZED ✓
- Blog post PUBLISHED ✓
- 50+ commits this phase

DSA IN JAVA:

- 50+ problems SOLVED ✓
- All basic data structures ✓
- Interview CONFIDENT ✓

WEB DEVELOPMENT:

- Portfolio website LIVE ✓
- Responsive & beautiful ✓
- Showcases projects ✓

PYTHON:

- Expert level ✓
- Data transformation ✓
- Advanced concepts ✓

GITHUB:

- 61 daily commits ✓
- Professional portfolio ✓
- Impressive graph ✓

STATISTICS:

- Total days: 61 
- Total hours: 366 
- DSA problems: 50+ 
- Commits: 61 
- Blog posts: 1 
- Projects deployed: 1 

FINAL STATUS:  CHAMPION LEVEL!

 COMPARISON: JUNE vs JULY

Metric	JUNE (Phase 1)	JULY (Phase 2)	Total
Days	30	31	61
Hours	180	186	366
DSA Problems	0→30	30→50+	50+
Data Eng	Learning	Production	Deployed
Web Dev	Building	Live	Live
Blog Posts	0	1	1
Commits	30	31	61
Status	Foundation	Complete	Champion

 SUCCESS PATTERNS

JUNE Pattern:

Week 1: Setup & Learn
Week 2: Fundamentals solid
Week 3: First project
Week 4: Polish & prepare

Result: Ready for deployment

JULY Pattern:

Week 5: Deploy to cloud

Week 6: Optimize & master

Week 7: Document & blog

Week 8: Final polish

Result: Interview-ready portfolio

⚠️ CRITICAL SUCCESS FACTORS

Phase 1 (June) Critical Success:

1. **Never miss a day** - Consistency = Foundation
2. **Complete Spark pipeline** - Your main project
3. **Reach 30 DSA problems** - Interview prep
4. **Daily commits** - Show progress

Phase 2 (July) Critical Success:

1. **Deploy everything** - Proof of capability
2. **Publish blog post** - Demonstrate communication
3. **Reach 50 DSA problems** - Expert level
4. **Polish everything** - Professional appearance

🎉 AUGUST 1 FINAL CHECKLIST



DATA ENGINEERING

- ☐ Spark pipeline deployed on AWS ☒
- ☐ Docker container production-ready ☒
- ☐ Blog post published ☒
- ☐ Code well-documented ☒
- ☐ GitHub professional ☒



DSA IN JAVA

- ☐ 50+ problems solved ☒
- ☐ All major data structures ☒
- ☐ Interview-confident ☒
- ☐ Time/space complexity mastered ☒



WEB DEVELOPMENT

- ☐ Portfolio live on GitHub Pages ☒

- ☐ Responsive design 
- ☐ Showcases projects 
- ☐ Professional appearance 

PYTHON

- ☐ Expert-level fundamentals 
- ☐ Data transformation 
- ☐ OOP concepts clear 
- ☐ Production-ready code 

GITHUB

- ☐ 61 daily commits 
- ☐ Green commit graph 
- ☐ Professional portfolio 

DOCUMENTATION

- ☐ README complete 
- ☐ Setup guide ready 
- ☐ Blog post published 
- ☐ Architecture documented 

FINAL VERDICT:  READY FOR INTERNSHIP INTERVIEWS!

WHAT COMES NEXT (After August 1)

Your options with your new skills:

1. Internship Applications

- Amazon, Google, Microsoft, flipkart...
- Startups
- Data engineering firms

2. Full-Stack Learning

- Web development mastery
- Full-stack projects
- System design

3. Specialization

- Deep ML/AI
- DevOps/Cloud
- Advanced Data Engineering

You've got 61 days. Make them count! 